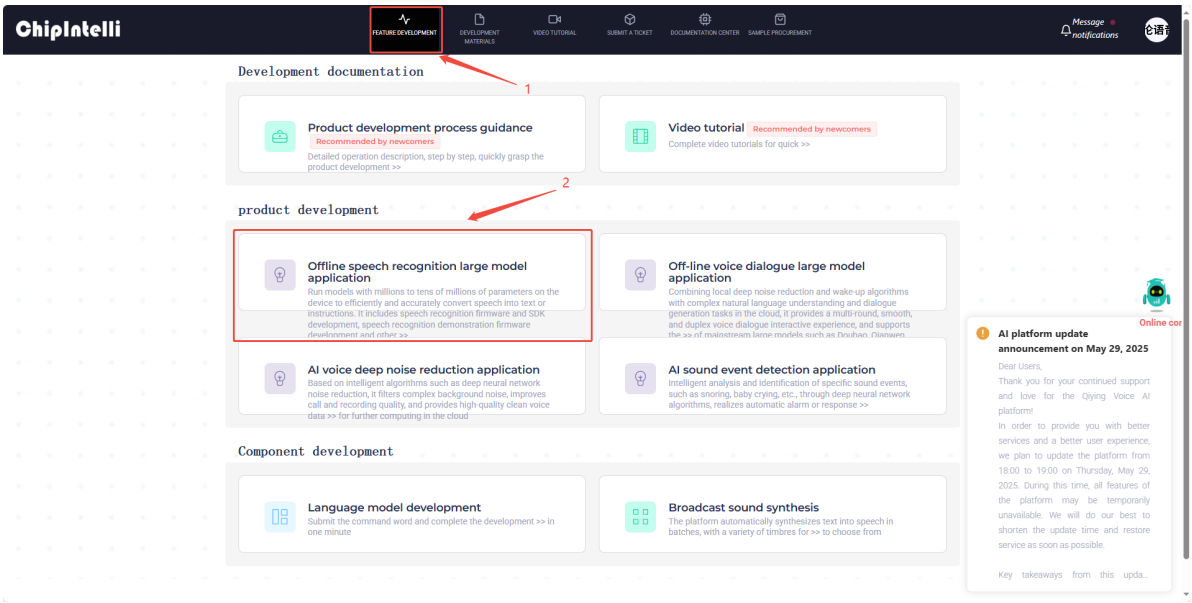
Modify the wake-up word and command word

- First, you need to open the link "[ChipIntelli Voice AI Platform]" and enter the official firmware production website.
- Translate this page into English.

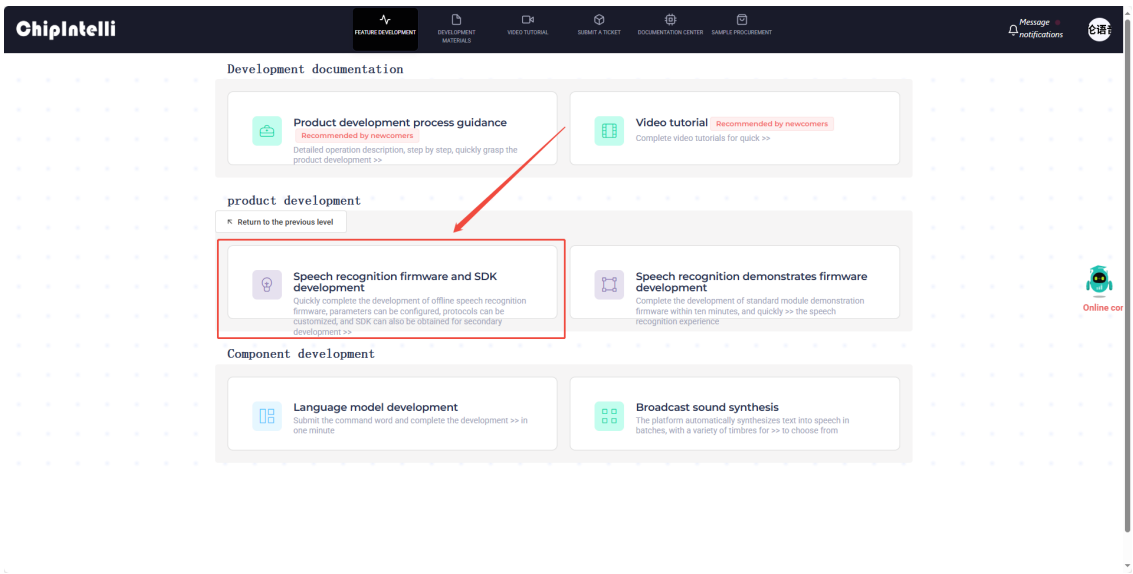
- Click **"Function development"** in the menu bar, and then click **"Offline speech recognition large model application"** under the Product Development column.



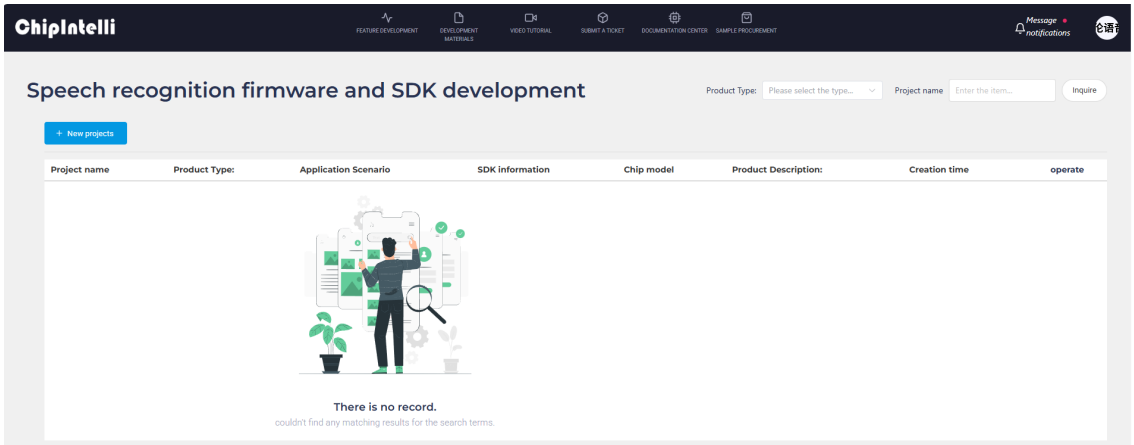
- At this time, you will be prompted to log in. Here you need to use your own information to register a platform account. The tutorial here has already been registered in advance.

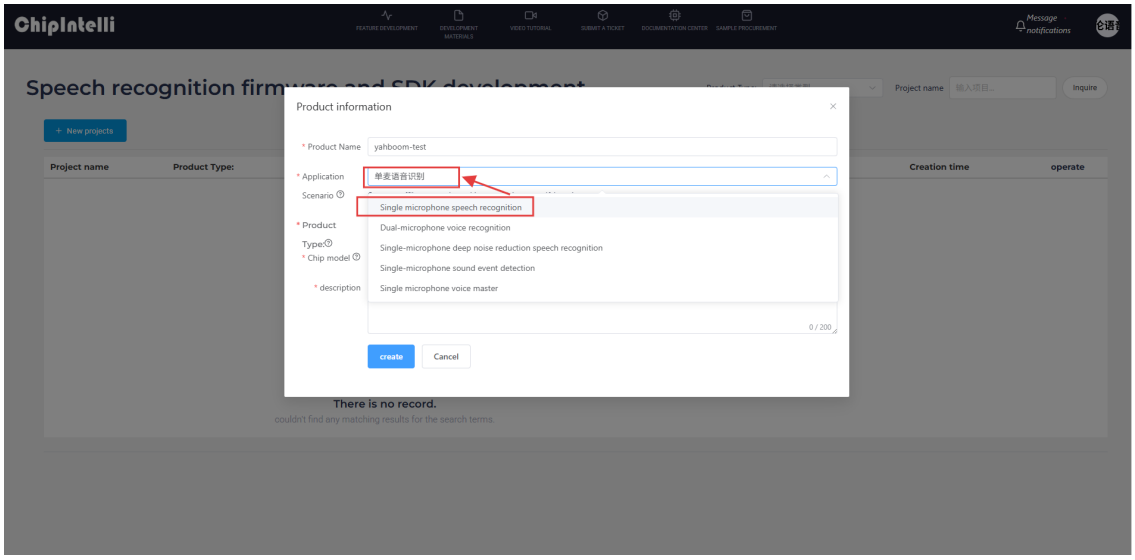
```
If you don't have a Chinese mobile phone number, you can use the following
account:
Account number: 15338857526
Password: Yahboom123

Note: Since multiple people may share one account, please be careful not to
modify other people's firmware when using it.
```

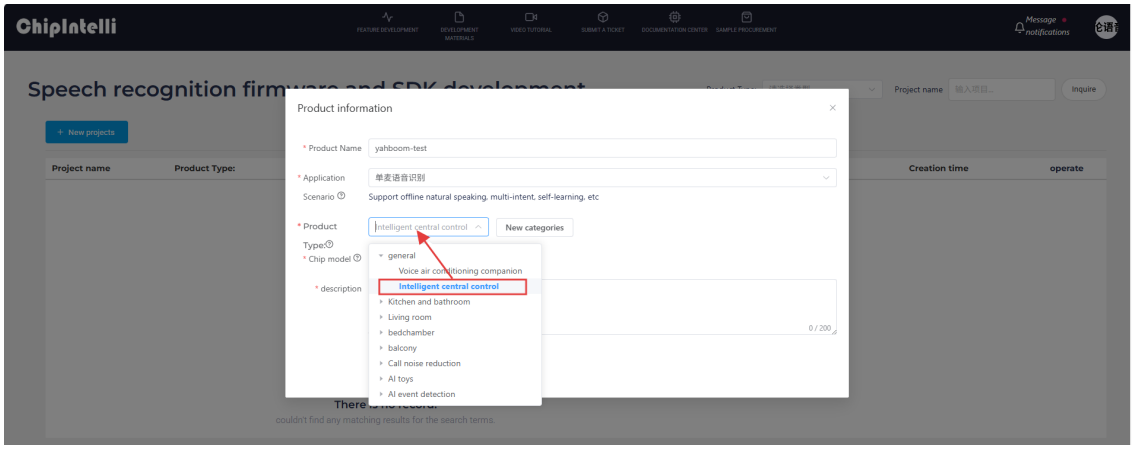
After logging in, click **“Speech recognition firmware and SDK development”** again.

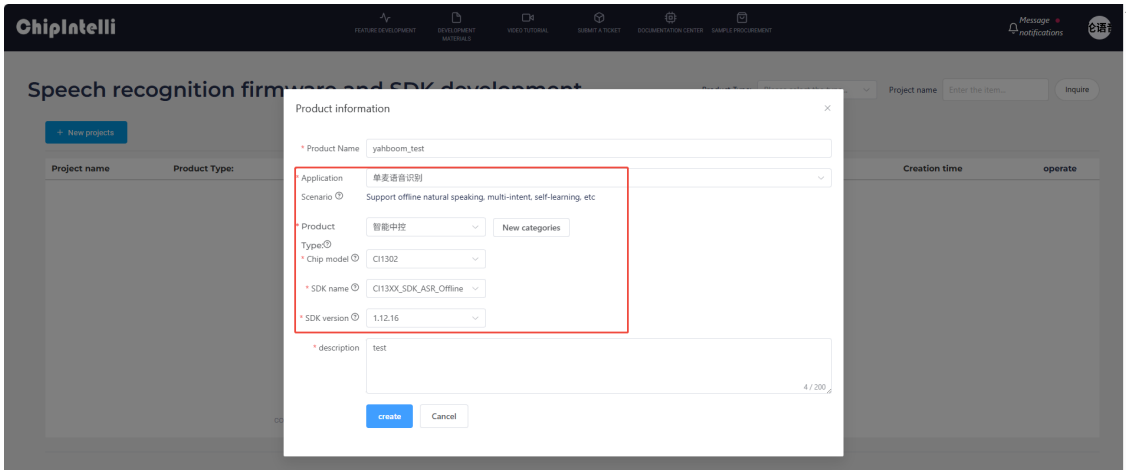


- After the page jumps, click New Project on the left and create a new product as shown below. The product name and description can be customized. The rest of the information needs to be selected according to the content in the red box. The product model needs to be selected as **"General->Intelligent central control"**. After completion, click Create

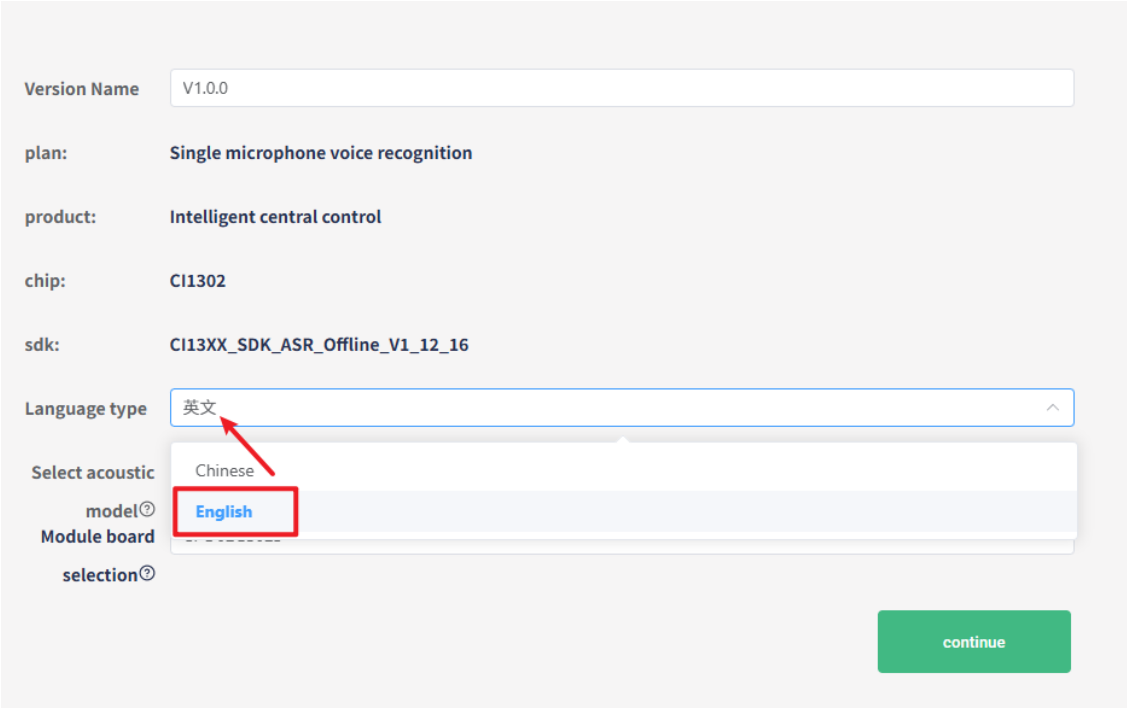


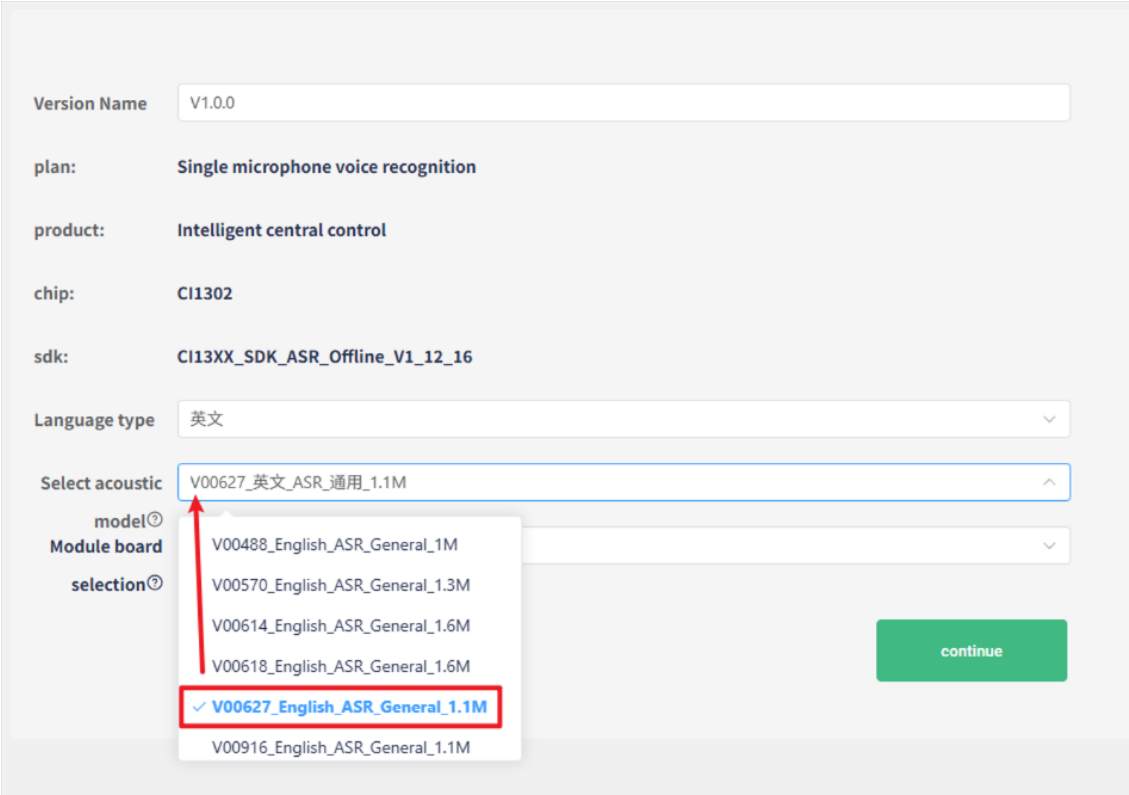


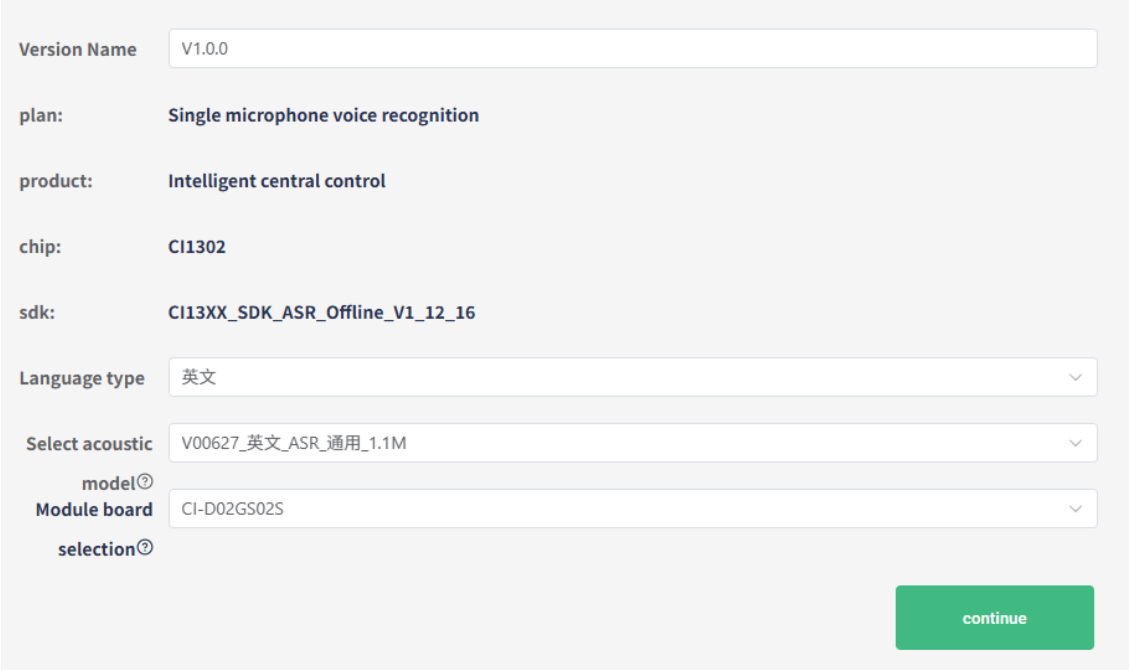




- Next, you need to fill in the basic information of the project. We need to recognize English, so select "English" as the language type. If you need to recognize Chinese, you can also make corresponding changes. For other information, select as shown below. Click Continue when you are done.







- Then you need to configure the firmware. Here we only explain the parts that need to be modified and turn on the echo cancellation algorithm parameters.

Algorithm parameters	parameter	illustrate	Options
	Echo cancellation	This is suitable for scenarios that require voice interruption (such as music playback), and you need to confirm that the module hardware supports this function before you can enable it.	close open

- In the hardware parameters, you need to select the crystal source as "**internal RC**".

Hardware parameters	parameter	illustrate	Options
	Core 1.1V power supply	External DC-DC or internal LDO power supply	内部
	Crystal oscillator	Internal RC or external crystal source is optional	内部RC
	parameter	illustrate	Internal RC External crystal oscillator
	baud rate calibration function	Use baud rate calibration function when using internal RC.	

- In the print serial port configuration, configure the UART1 level as open drain function, supporting external pull-up 5V.

Print serial port configuration	parameter	illustrate	Options
	Debugging information printing serial port	During debugging, the serial port (baud rate 921600) that prints debugging information must not be the same as the communication serial port; if not needed, simply turn it off.	UART1
	parameter	illustrate	Options
	UART1 level	Configured with open-drain pins, supporting external 5V pull-up.	close open

- Modify the communication serial port configuration, set the baud rate to 115200, and configure the UART0 level to open drain function, support external pull-up 5V, and click "**Continue**" to enter the next step after the configuration is completed.

Communication serial port configuration	parameter	illustrate	Options
	Serial communication	When enabled, the voice chip (module) communicates with the host computer via a designated serial port; when disabled, it cannot communicate with the host computer.	close open
	Communication serial port	Serial port name for communication	UART0
	Communication serial port baud rate	baud rate of serial communication	115200
	Serial port protocol version	Custom serial communication protocols consist of up to 20 hexadecimal digits, such as: AF BF D1 01 12 FB. When the voice chip receives protocol packets, it determines that the packet has been received by measuring the reception timeout duration (0.34ms). Therefore, when the main controller sends protocol packets, it needs to ensure that the time interval between packets is greater than 0.34ms to ensure normal communication.	自定义协议
	parameter	illustrate	Options
	UART0 level	Configured with open-drain pins, supporting external 5V pull-up.	close open

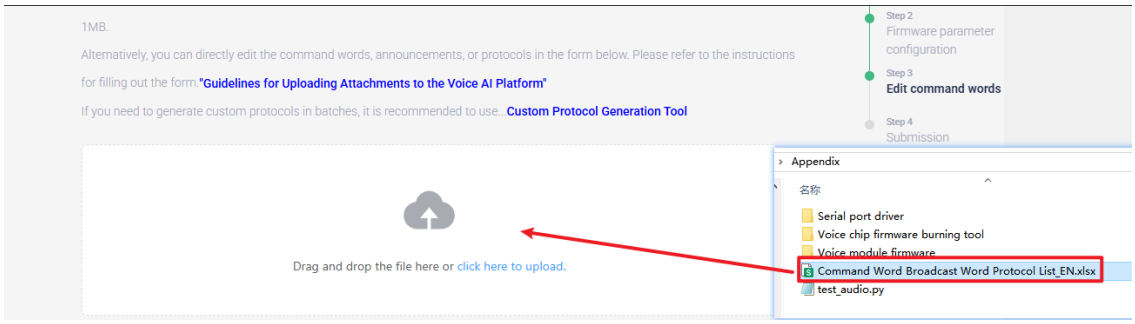
- Next, enter the edit command word function. First, you need to select the sound to be played.

tone selection

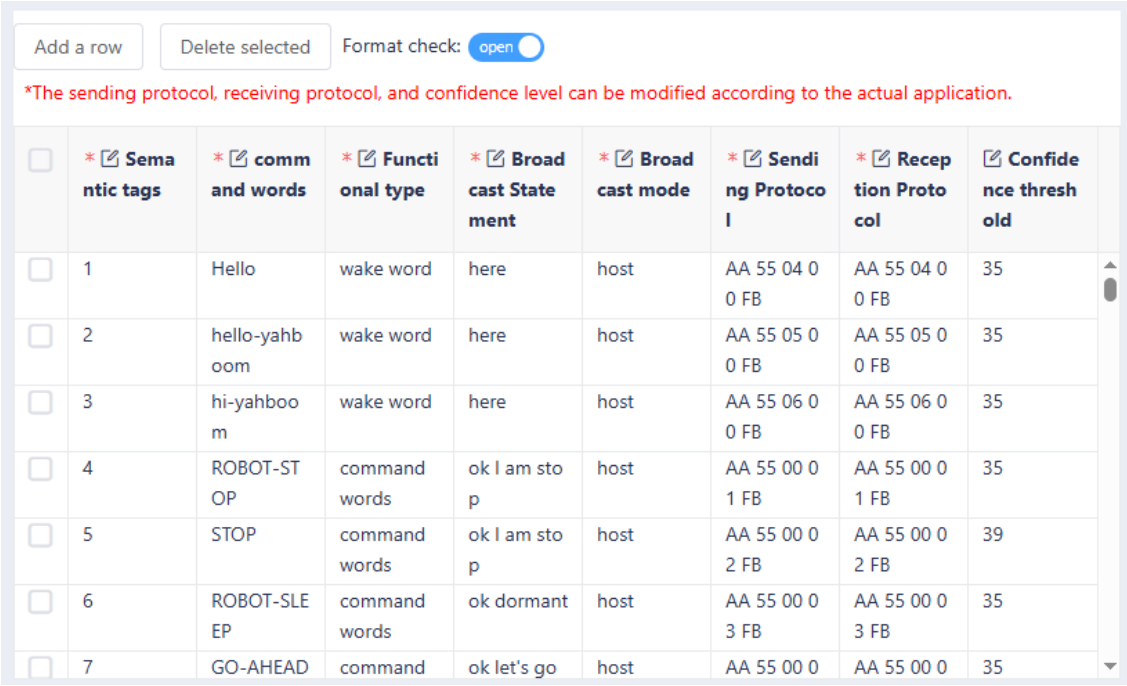
English - Adult Female Voice | English - Adult Male Voice | English - Girl's Voice

Ana - English Female Voice Ver.3
Olivia - English Female Voice Ver. 3
Sophia - English Female Voice Ver. 3
Mia - English Female Voice Ver. 3
Harper - English Female Voice Ver. 3
Linda - English Female Voice Ver. 3
Dora - English Female Voice Ver.2
Rebecca - English Female Voice Ver.2

- Next, we will upload the command word attachment. Locate the protocol table file "Command Word Broadcast Word Protocol List_EN.xlsx" in the download section and drag and drop it directly into the webpage to upload it.



- After uploading the file, you can see our command word data in the table below.

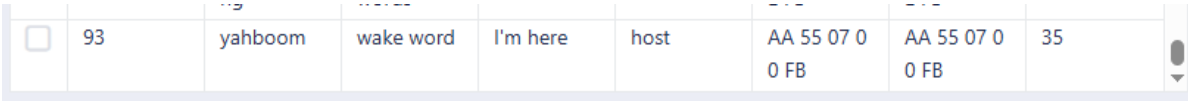


Add a row | Delete selected | Format check: open

*The sending protocol, receiving protocol, and confidence level can be modified according to the actual application.

* Semantic tags	* command words	* Functional type	* Broadcast Statement	* Broadcast mode	* Sending Protocol	* Reception Protocol	Confidence threshold
1	Hello	wake word	here	host	AA 55 04 00 FB	AA 55 04 00 FB	35
2	hello-yahboom	wake word	here	host	AA 55 05 00 FB	AA 55 05 00 FB	35
3	hi-yahboom	wake word	here	host	AA 55 06 00 FB	AA 55 06 00 FB	35
4	ROBOT-STOP	command words	ok I am stop	host	AA 55 00 01 FB	AA 55 00 01 FB	35
5	STOP	command words	ok I am stop	host	AA 55 00 02 FB	AA 55 00 02 FB	39
6	ROBOT-SLEEP	command words	ok dormant	host	AA 55 00 03 FB	AA 55 00 03 FB	35
7	GO-AHEAD	command	ok let's go	host	AA 55 00 0	AA 55 00 0	35

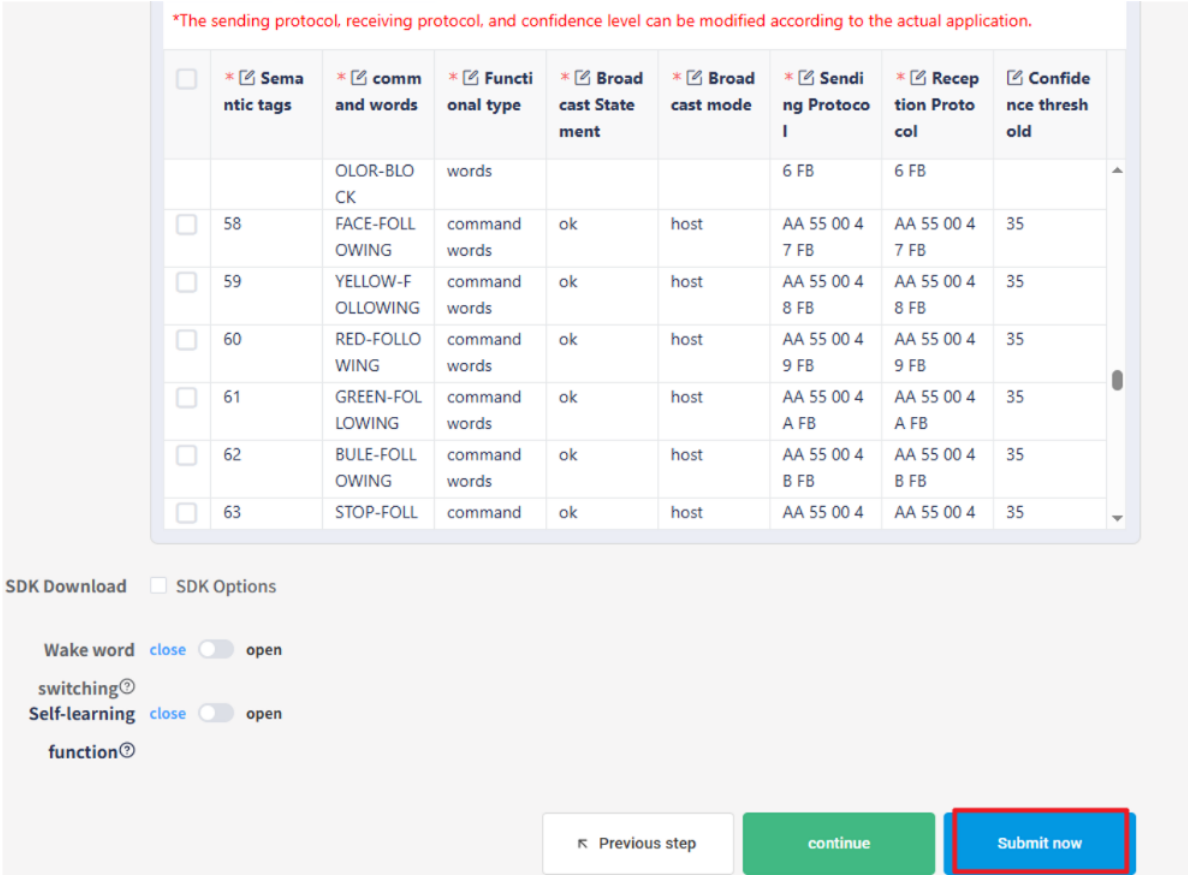
If we want to add a wake word, we click **Add a row**. If we need to add a new wake word "yahboom", then add the following content to the new line:



93	yahboom	wake word	I'm here	host	AA 55 07 00 FB	AA 55 07 00 FB	35

The sequence **AA 55 07 00 FB** is incremented from the previous command word "hi-yahboom" **AA 55 06 00 FB**. Subsequent commands follow the same pattern; therefore, the next wake word's send/receive protocol would be AA 55 08 00 FB. A maximum of 10 wake words can be set.

- Click to submit now



*The sending protocol, receiving protocol, and confidence level can be modified according to the actual application.

	* Semantic tags	* command words	* Functional type	* Broadcast Statement	* Broadcast mode	* Sending Protocol	* Reception Protocol	Confidence threshold
		OLOR-BLOCK	words			6 FB	6 FB	
	58	FACE-FOLLOWING	command words	ok	host	AA 55 00 47 FB	AA 55 00 47 FB	35
	59	YELLOW-FOLLOWING	command words	ok	host	AA 55 00 48 FB	AA 55 00 48 FB	35
	60	RED-FOLLOWING	command words	ok	host	AA 55 00 49 FB	AA 55 00 49 FB	35
	61	GREEN-FOLLOWING	command words	ok	host	AA 55 00 4A FB	AA 55 00 4A FB	35
	62	BULE-FOLLOWING	command words	ok	host	AA 55 00 4B FB	AA 55 00 4B FB	35
	63	STOP-FOLL	command	ok	host	AA 55 00 4	AA 55 00 4	35

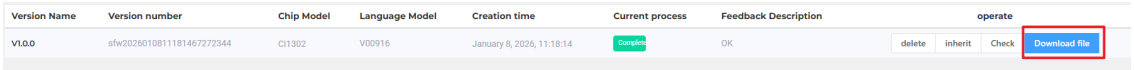
SDK Download SDK Options

Wake word switching close open

Self-learning function close open

Previous step continue Submit now

- Wait for firmware generation, which takes about 3 minutes. When the **Download File** display turns blue, you can download the firmware.



Version Name	Version number	Chip Model	Language Model	Creation time	Current process	Feedback Description	operate
V1.0.0	sfw20260108111814672723 44	CI1302	V00916	January 8, 2026, 11:18:14	Complete	OK	delete inherit Check Download file

- Next, flash the firmware. Refer to section 4.3 in [10.AI Large Model Voice Module (Must Read)] - [2. Flash the firmware (Must read)]. After flashing is complete, follow the tutorial to start [11.2D Voice Control Course] - [1.Simple Voice Control]. After the program starts, use the wake word "yahboom" to wake up the device. If it responds "I'm here", the addition was successful.
- If adding a command word, change the function type to command word, then modify the send protocol and receive protocol. For example, if we want to add a command word **arm-dancing**, we can add the following content to the **new row**:

				ment		l	col	old
	89	Tracking-and-picking-up-objects	command words	ok	host	AA 55 00 6B FB	AA 55 00 6B FB	35
	90	Start-garbage-sorting	command words	ok	host	AA 55 00 6C FB	AA 55 00 6C FB	35
	91	Reorder	command words	ok	host	AA 55 00 6D FB	AA 55 00 6D FB	35
	92	arm-centering	command words	ok	host	AA 55 00 6E FB	AA 55 00 6E FB	35
	93	yahboom	wake word	I'm here	host	AA 55 07 00 FB	AA 55 07 00 FB	35
	94	arm-dancing	command words	ok	host	AA 55 00 6F FB	AA 55 00 6F FB	35

The subsequent send/receive protocol is incremented from the command word "arm-centering" on line 92, whose protocol is AA 55 00 6E FB. We incremented 6E to 6F. By analogy, the following ones are 7A, 7B, etc.

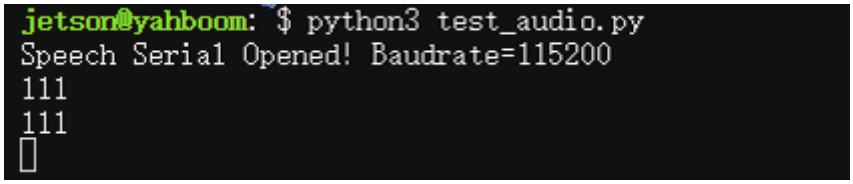
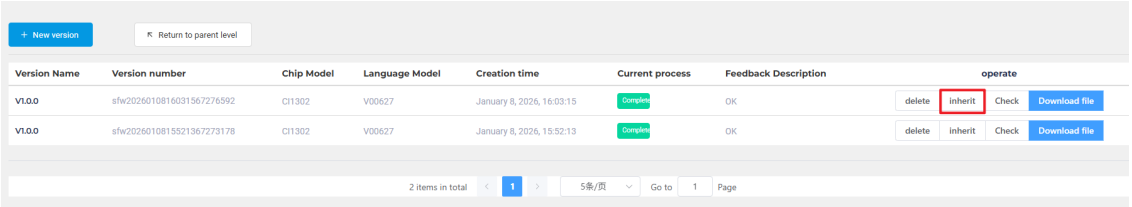
- Similarly, after submission, wait for the firmware to be created and then download it. Then, burn it into the audio board. Next, connect the board to the motherboard and copy the `test_audio.py` file from the program source code folder to the `/root` directory of Docker. First, place `test_audio.py` in the `~` directory of the host machine, then enter the following command to copy the file into Docker and test the added command.

```
docker cp ~/test_audio.py <container_id>:/root
```

Then use `docker exec` to enter the container, and then enter the following in the container's terminal:

```
python ~/test_audio.py
```

Say "Hello yahboom" to wake up the voice module. The module will respond "I'm here", indicating successful wake-up. Then say "Arm dancing" to the voice module. If the terminal prints 111, it means recognition was successful. Here, 111 corresponds to the decimal value of 6F.



- Whether updating command words or wake words, it is recommended to select "inherit" the previous configuration each time.